Session-V: Estimation of groundwater recharge: September 3rd week

Program schedule

S.No	Time	Content	Method	Materials	Expected outcomes
1	15 min.	Recap.	Large group discussion	Pictures, charts and markers.	Participants recollect the contents discussed in fourth session.
2	15 min.	Discussion on homework: Estimation of total rainfall received (HU wise consolidated data).	Large group discussion	Charts and markers.	Participants estimate total rainfall received by consolidating the data from all the rain gauge stations in the HU.
3	45 min.	Estimation of HU area.	Large group discussion and small group discussion	Graph sheets, carbon papers toposheets, pencils, scale, calculator, sketch pens, charts, pins and markers.	Participants experience the steps involved in estimation of Hydrological Unit area.
4	45 min.	Estimation of groundwater recharge (June-October).	Large group discussion, and small group discussion	Rainfall data, charts, markers and calculators.	Participants identify the information need to estimate groundwater recharge and estimate the recharge.
5	20 min.	Group dynamics-Data accuracy.	Demonstration	Three glasses, one transparent bowl, colour powder and water.	Participants will demonstrate the importance of data accuracy in estimating groundwater levels.
6	20 min	Short Study: Weed management.	Demonstration	Bottles, weed plants, rice plant, blade, kesar colour, charts and markers.	Participants discuss the need for weed management.
7	30 min.	Homework: Data collection for estimation of groundwater draft Rabi crop plans, and secondary water bodies.	Large group discussion	Charts and markers.	Participants take part in data collection, discussion for estimation of groundwater draft Rabi crop plans, and secondary water bodies.
8	10 min.	Evaluation.	Large group discussion	Charts and markers.	Participants share their opinions on the conduct of FWS sessions and methods used in the session.

Learning objectives

By the end of the session, participants will be able to:

- Recap the previous session,
- Estimate total rainfall received,
- Estimate the HU area,
- Estimate the groundwater recharge,
- Understand the importance of data accuracy,
- Analyze the water level fluctuations (SWL, PWL Vs rainfall), and
- Discuss the data required for estimation of draft and explore the ways to collect the data.

Process

1. Recap Time: 15 min.

Objective : Participants recap the previous session.

Content :

- Natural and artificial recharge,
- Factors Influencing recharge (HESA),
- Recharge rates (GEC norms) of different rock types,
- Importance of cooperation among GMC members, and
- Homework- Collection of rainfall data (June-October).

Method : Large group discussion

Materials : Charts and markers.

Process

- Welcome the participants to the fifth session.
- In the large group, encourage each participant to share a content that was discussed in previous session and the learning from that one.
- As you facilitate the recap, ask one of the participants to list the contents on the chart paper.
- Discuss the contents and the learning from the previous session.

2. Discussion on Homework – Estimation of total rainfall received Time: 15 min.

Objective : Participants estimate total rainfall received by consolidating the data from all the rain gauge stations in the HU.

Content : Estimation of total rainfall in the HU.

Method : Large group discussion

Materials : Charts and markers.

Process

- Encourage the participants to fill in the rainfall data on the chart. Use the following chart as a reference:

S.No.	Name of the rain gauge station	June	July	Aug.	Sept.	Oct.	Total
1							
2							
3							
4							

Process of taking averages:

- Encourage the participants to calculate the average rainfall of all the rain gauge stations of the HU.
- Share that we may not receive uniform rainfall throughout our hydrological unit. The average rainfall will give the picture about the rainfall received in the entire HU. So it is useful to calculate average rainfall received at all the rain gauge stations in the HU and consolidate the data month wise.
- Demonstrate the process of taking the 'average' for each month.
    - Take four empty one litre bottles and name each bottle with one rain gauge station and the month (June).
    - Fill each bottle with water according to the rainfall received by that rain gauge station for June month.
    - Transfer the water from four bottles into a big tumbler.
    - Again fill the bottles equally using the water in tumbler.
- Explain that each bottle represents the average rainfall of the Hydrological Unit for the month of June.
- Encourage participants to make averages for the other months (Jul, Aug, Sept and Oct).
- Calculate the average rainfall received for the period June to October.

3. Estimation of HU area Time: 45 min.

Objective : Participants estimate the area of the Hydrological Unit.

Content : Area of Hydrological Unit.

Method : Large group discussion, and small group discussion.

Materials : Graph sheets, carbon papers, toposheet, pencils, scale, calculator, sketch pens, charts, pins and markers.

Process

- Remind participants that the area of the HU is critical factor in estimation of groundwater recharge. To facilitate the process of estimation of HU area, request participants to work in small groups.
- Facilitation should start with a demonstration.
- As part of demonstration, take HU map and place a tracing paper on the map.
- While observing the HU map through the tracing paper, sketch out the map outline.

- Place the tracing paper on a graph and trace the map outlines so as to appear on the graph sheet.
- Colour the full squares with red, ¾ squares with yellow, ½ squares with green and ¼ squares with blue.
- Next, count the, ¼ squares, ½ squares, ¾ squares and full squares on the graph sheet.
- Convert the part squares (¼, ½, ¾ squares) into full squares to calculate the area easily. For example: four ¼ squares make one full square; two ½ squares make one full square; two ¾ squares make 1½ squares; or four ¾ squares make 3 squares.
- Considering the scale in toposheet (1: 50,000) area of a one full square = 62.5 acres or 25 hectares. Area of HU = No. of full squares X Area of one full square.
- Encourage Participants to work in small groups and calculate the area of Hydrological Unit. Provide assistance to the groups in the process of estimation.

4. Estimation of groundwater recharge Time: 45 min.

Objective : Participants estimate the groundwater recharge through rainfall.

Content

- Average rainfall received during June – October.
- Recharge rates of different rock types.
- Estimation of groundwater recharge.

Method : Large group discussion, and small group discussion

Materials : Rainfall data, charts, markers and calculators.

Process

- Remind the participants about the amount of average rainfall received.
- Share the amount of average rainfall received once again and discuss the process of estimation of recharge.
- Ask participants the following question: "What information is required to estimate recharge of the particular Hydrological Unit?"
- List participant's responses on a chart.
- The data required to estimate the groundwater recharge is:
    - Amount of average rainfall received in that HU from June to October.
    - Area of the HU.
    - Type of rocks and it's extent.
    - Recharge rates (GEC Norms).
- Next, encourage participants to share about the rock formations and their extent. Based on the type of rock, share the recharge rate of that particular rock type and estimate the groundwater recharge of the HU.

Recharge rate (GEC norms) in different rock types .

- Begin the discussion with these words: "Now we know average rainfall received during June to October, Let us think about rate of percolation". Ask participants the following question: "Do you think rate of water percolation is same in all types of rocks?"
- Write participants responses on a chart paper and list the different kinds of rocks.

- Next, pose the question: "Do you think all the rain water that falls, percolates the ground?"
- Listen to the participants responses and encourage them to demonstrate the following experiment:

Experiment:

- Take a dry sponge.
- Pour 500 ml. of water on sponge similar to rainfall.
- Collect water into a tray by squeezing the sponge.
- Observe that less water is collected into the tray.
- Following the discussion, summarize as follows: "Rain water flows through streams, rivulets and fills the tanks, ponds before the streams join the river and then to sea. A small portion of the rain water infiltrates into the ground."
- Explain to farmers that all the rainwater does not infiltrate into the soil. Rate of infiltration depends upon rock formation, and soil structure.
- Share the recharge rates recommended by Ground Water Estimation Committee (GEC) with the participants.

Rock type	**Infiltration rate**
a. Alluvial areas	
Indo-Gangetic and inland areas	22%
East coast	16%
West coast	10%
b. Hard rock areas	
Weathered granite, gneiss and schist with low clay content	11%
Weathered granite, gneiss and schist with significant clay content	8%
Granulite facies like charnockite etc.	5%
Vesicular and jointed basalt	13%
Weathered basalt	7%
Laterite	7%
Semi consolidated sandstone	12%
Consolidated sandstone, Quartzites, Limestone (except cavernous limestone)	6%
Phyllites, Shales	4%
Massive poorly fractured rock	1%

- Next, before estimation of groundwater recharge discuss about the measuring unit of water flow and groundwater recharge and explain the process of conversion with the following example.

Cubic meter

Share the following information with the participants**: "**Usually, we measure rainfall in millimeters. But, the area of the HU is in Square meters. So, In order to bring all the parameters into the same

type of unit and enable easy calculation, we need to convert one unit to the equivalent of the remaining.

Cubic meter is the common measuring unit for calculating water flow and recharge. Farmers estimate the amount of recharge in liters. Liters being a small measuring unit, quantification of the estimate in liters run into large numbers/figures.

Apart from reducing the numbers, as one cubic meter equals 1000 liters, it is a common measuring unit for estimation of recharge and draft. Helping farmers understand quantification of water in cubic meters will enable them to interact with other stakeholders in groundwater management".

Let us calculate the quantity of water received in an area of one acre for 25 millimeters (mm) of rainfall.

1. Convert the rainfall into metres.

25 mm = 25/1000

= 0.025 metres.

2. Convert the area into square metres (Sq.m).

One acre = 4000 sq.m.

3. Multiplication of area and rainfall will give the amount of water received in an area of one acre.

Area (sq.m) x rainfall (m) = Cubic metres (cu.m).

4000 x 0.025 = 100 cu.m.

One cubic metre = 1000 litres.

100 cu.m = 100 x 1000

= 100000 litres.

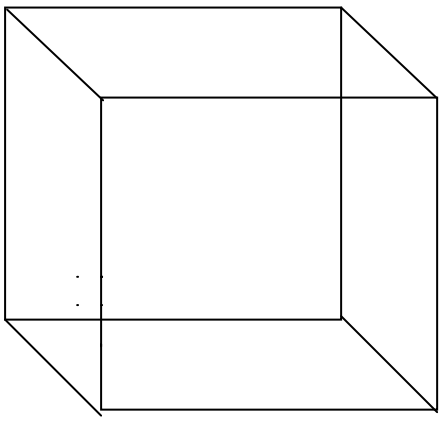
4. Explain what a cubic metre is:

Any object with the specifications of

1 metre long, 1 metre wide, and

1 metre height is called a cubic metre.

[(1m (L) x 1m (W) x 1m (H)]



5. We are all aware that all the rainwater does not infiltrate into the ground. A small portion of it (i.e. < 10%) infiltrates into the ground.

So the quantity of water infiltrated into the ground = 100000 x 4/100

= 4000 litres.

- Divide the participants into five groups.
- Assign each group one month's rainfall and ask the groups to estimate groundwater recharge.
- Visit each group and provide necessary assistance.
- Ask the small groups to report their estimation results to the large group.

- Calculate the total recharge by adding the recharge of each month.

5. Group dynamic: Data accuracy Time: 20 min.

Objective : Participants will demonstrate the importance of data accuracy in estimating groundwater levels.

Materials : Three glasses, one transparent bowl, colour powder and water.

Method : Game

Process

- Select three volunteers and give one transparent glass to each volunteer.
- Ask two volunteers to get some water and pour into the transparent bowl.
- Then, ask the third volunteer to bring coloured water and pour into the transparent bowl.
- Ask the participants to observe the colour of water in the transparent bowl after the third volunteer pours in.
- Discuss the water that was brought by first two volunteers with correct information and water brought by third volunteer with misinformation.
- Conclude that correct information is very critical for any activity, especially for Crop Water Budgeting exercise.

6. Short study: Weed management Time: 20 min.

Objective : Participants discuss the need for weed management.

Rationale : Weeds generally compete for the nutrients and water with the main crop. Conduct of this experiment would help participants understand the importance of weed management.

Materials : Three 100 ml. bottles, two types of weed plants, rice plant, blade, Kesar colour, one litre water, markers and charts.

Usage

- Ask the participants to bring two types of weed plants of the same height and a rice plant along with roots.
- Wash the root zone thoroughly and cut the ends of the roots with an angle.
- Fill the three bottles with 90 ml. of coloured water. Wrap cotton around and insert the plants into the bottles to avoid entry of air into the bottles.
- Write the names of the plants on the bottles.
- Observe the results after one hour.
- Discuss the competition of weeds for nutrients and water and the need for weed management.
- Next, encourage participants to discuss ways of weed management. Invite a few participants to share their experiences.

7. Homework: Data collection for estimation of groundwater draft & Rabi crop plans, secondary water bodies. Time: 30 min.

Objective : Participants discuss:

- Data required for estimation of draft and the ways of collection.
- Need to collect Rabi crop plans.
- Ways to collect the data on projected rainfall and secondary water bodies.

Content :

- Data required for estimation of draft during June – October,
- Rabi crop plans,
- Secondary water bodies, and
- Ways of collection of data.

Method : Large group discussion

Materials : Charts, and markers.

Process

- Begin the discussion with: "so far we have discussed groundwater recharge."
- Ask participants how we are spending this groundwater.
- Then ask the following questions in large group.
  a. What is the data required to estimate the draft?
  b. How can we collect the required data?

Tips to facilitator

- The following data is required for estimating draft:
  - Number of functioning bore wells,
  - Number of functional days of each bore well,
  - Number of functional hours on each day, and
  - Drum discharge (litres per hour).
- It is better to collect number of bore wells connected to each transformer.
- Number of functional days of each bore well and functional hours in a day should be collected from the owner of the bore wells.
- Drum discharge of an observation bore well in that village.

- Next discuss the need for Rabi crop plans to estimate draft for Rabi season.
- Encourage the participants to discuss the ways to collect the Rabi crop plan details and the role of GMCs in data collection.

Tips to Facilitator:

Data required is:

- The crop and area cultivated under each bore well in the habitation.
- Involve GMCs in data collection.

- Next discuss the need for data on projected rainfall, and secondary water bodies to estimate groundwater balance by the end of May.
- Encourage participants to discuss ways to collect the data on projected rainfall and secondary water bodies.

Tips to Facilitator:

Encourage the participants to discuss:

- Projected rainfall: To estimate projected rainfall (Nov-May) we need to collect the past 10 year's rainfall data for the same period. It can be collected from nearest Tahsildar's office.
- Secondary water bodies: The following information is to be collected from each village:
    - No. of tanks, ponds, check dams;
    - Area of water in tanks, ponds and check dams;
    - Height of water column;
    - No. of fillings; and
    - No. of days water stored in the tanks, ponds, check dams.

8. Evaluation Time: 10 min.

Ask the following questions to evaluate the day's learning. Pose the questions to the large group:

- Why do we need to consolidate rainfall data?
- List the steps involved in the estimation of Hydrological unit area?
- Why do we need to convert the measuring units of recharge and area? And what is the common measuring unit for the amount of recharge?
- How do you plan to share this information in GMC/ HUN meetings?